

Sensor HW3

March 25, 2025

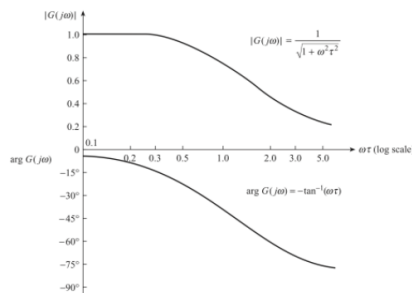
Please set up a light measurement circuit according to the figure below. Use an Arduino board to output a 5 Hz square wave to drive an LED. Then, build a light measurement system using a photoresistor and measure the voltage across it.

1. Download the data from the oscilloscope and fit it using a first-order system equation

$$V(t) = Ae^{-\frac{t}{\tau}} + B$$

to obtain the time constant τ for both the rising edge and the falling edge by data fitting. Why are these two time constants different?

2. Record the peak-to-peak voltage from the previous case. Then increase the square wave frequency to 50 Hz, 500 Hz, and 5000 Hz, and measure the peak-to-peak voltage for each case.
3. Use your knowledge of system transfer functions to explain your observations in part (2).



1. For 5 Hz square wave, we download the data from the oscilloscope and plot the whole waveshape in Fig.1:

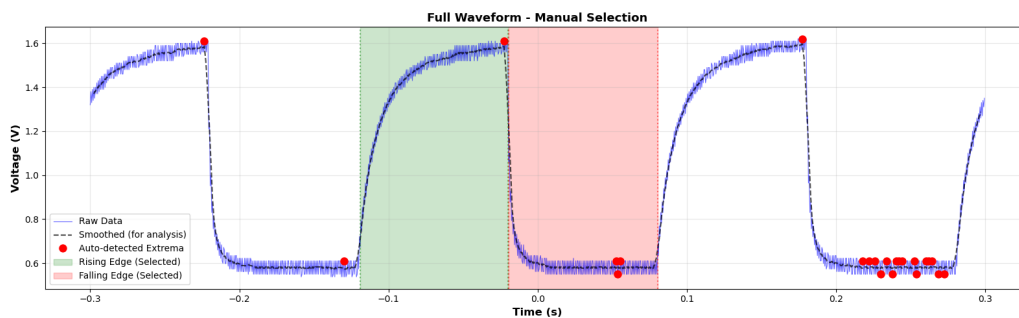


Figure 1: Wave form

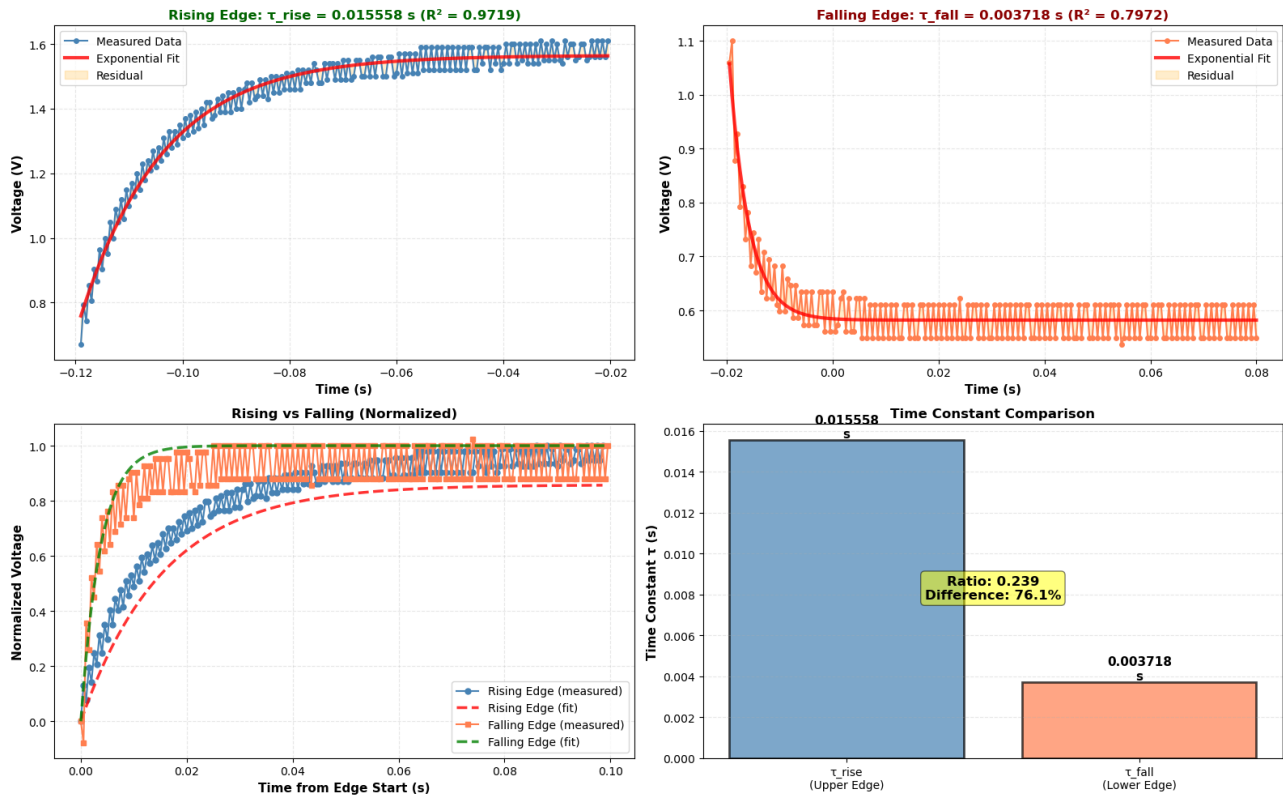


Figure 2: Curve fitting and analysis

We select data points 362 to 560 as the rising segment, and 561 to 761 as the falling segment. For rising phase, we use $V(t) = A(1 - e^{-\frac{t}{\tau}}) + B$ to fit the curve. For falling phase, we use $V(t) = Ae^{-\frac{t}{\tau}} + B$ to fit the curve.

From Fig.2 we know the time constant τ for rising edge is **0.015558s**, for falling edge is **0.003718s**.

3.

Frequency (Hz)	Minimum frequency	Maximum frequency	Peak-to-peak voltage
5	0.6	1.61	1.01
50	0.571	1.19	0.619
500	0.912	0.653	0.259
5000	0.57	0.768	0.218

Table 1: Summary by frequency

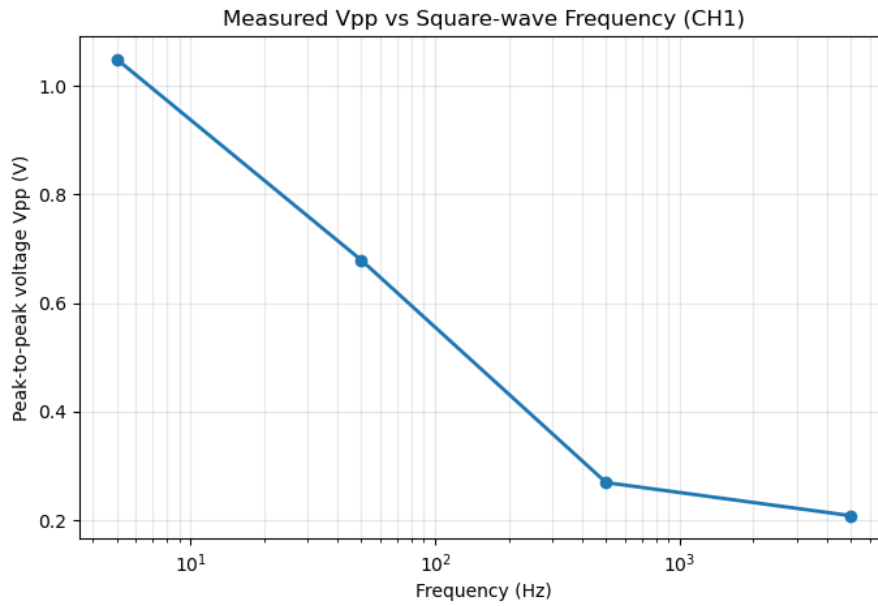


Figure 3: Vpp vs frequency

From the trend, we can tell that as frequency increases, Vpp drops compared with 5 Hz baseline. At sufficiently high frequency like 5000 Hz, output cannot fully charge/discharge within half-period.

An elastic force sensor has an effective seismic mass of 0.1 kg, a spring stiffness of 10 N m^{-1} and a damping constant of 14 N s m^{-1} .

- (a) Calculate the following quantities:
 - (i) sensor natural frequency
 - (ii) sensor damping ratio
 - (iii) transfer function relating displacement and force.
- (b) The above sensor is incorporated into a closed-loop, force balance accelerometer. The following components are also present:

Potentiometer displacement sensor: sensitivity 1.0 V m^{-1}
 Amplifier: voltage input, current output, sensitivity 40 A V^{-1}
 Coil and magnet: current input, force output, sensitivity 25 N A^{-1}
 Resistor: 250Ω .

 - (i) Draw a block diagram of the accelerometer.
 - (ii) Calculate the overall accelerometer transfer function.
 - (iii) Explain why the dynamic performance of the accelerometer is superior to that of the elastic sensor.

(a) $m = 0.1 \text{ kg}, k = 10 \text{ N} \cdot \text{m}^{-1}, c = 14 \text{ N} \cdot \text{s} \cdot \text{m}^{-1}$

$$\omega = \sqrt{\frac{k}{m}} = 10 [\text{rad/s}]$$

$$\zeta = \frac{c}{2\sqrt{km}} = 7$$

From Newton's second law,

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = \frac{k}{m}x_{\text{in}}$$

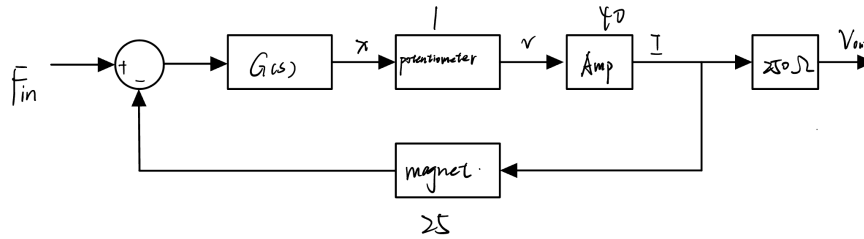
We apply Laplace's transform:

$$X(s^2 + \frac{c}{m}s + \frac{k}{m}) = \frac{F}{m}$$

Thus the transfer function is:

$$G(s) = \frac{X}{F} = \frac{1/m}{s^2 + \frac{c}{m}s + \frac{k}{m}} = \frac{10}{s^2 + 140s + 100}$$

(b)



$$G(s) = \frac{10}{s^2 + 140s + 100}$$

$$\frac{I}{F_{in}} = \frac{\frac{10 \cdot 40}{s^2 + 140s + 100}}{1 + \frac{25 \cdot 10 \cdot 40}{s^2 + 140s + 100}} = \frac{400}{s^2 + 140s + 10100}$$

Thus the overall accelerometer transfer function is:

$$\frac{V_{out}}{F_{in}} = \frac{400}{s^2 + 140s + 10100} \cdot 250 = \frac{10^5}{s^2 + 140s + 10100}$$

(iii) In an open-loop system, accuracy depends on the mechanical linearity of the spring, which is susceptible to temperature drift and aging.